5.5.2 Deciles

Deciles are similar to quartiles. Quartiles divides ungrouped data into four quarters and Deciles divide data into 10 equal parts .

Example 5.33

Find the D_6 for the following data

11, 25, 20, 15, 24, 28, 19, 21

Solution:

Arrange in an ascending order

11,15,19,20,21,24,25,28

$$D_6 = \left(\frac{6(n+1)}{10}\right)^{th} \text{ item}$$

$$= \left(\frac{6(8+1)}{10}\right)^{th} \text{ item}$$

$$= \left(\frac{6(9)}{10}\right)^{th} \text{ item}$$

$$= [5.4]^{th} \text{ item}$$

$$= 5^{th} \text{ item} + (0.4)\ (6^{th} \text{ item} - 5^{th} \text{ item})$$

$$D_6 = 21+(0.4)(24-21)$$

$$= 21+(0.4)(3)$$

$$= 21+1.2$$

$$= 22.2$$

Example 5.34

Calculate D_5 for the frequency distribution of monthly income of workers in a factory

Income (in thousands)	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24	24 – 28	28 – 32
No of persons	10	12	8	7	5	8	4	6

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